

- A. These are the items to focus on in addition to using your LECTURE NOTES as a study tool.
- B. I also created a Review Picture file. These are the figures that I would primarily pull from first when creating a picture/figure-based question on the exam. Remember, a single picture can have many points conveyed about a particular process or topic.
- C. This course is not simply memorizing and regurgitating information. You should be able to synthesize concepts we discussed and learn how processes work together.
- D. EXAM DATES/TIMES: >>>>> **SEE OUR SYLLABUS or MOODLE FOR DATE/TIME**
- E. You **MUST** take the exam during the announced time. Failure to do so will result in a zero on the exam, **UNLESS** you have made prior arrangements with me. There are no exceptions to this policy.
- F. EXAM will be taken at the LSU Testing Center without the use of **ANY** outside materials or resources. This is the only format that the exam is offered.
- G. **Missed exams will result in a zero (0%), no exceptions.** Do NOT miss your exam session.
- H. You can also go into Mastering Biology and create practice exams and review materials to see how well you understand the material prior to taking our exam. Please see the video posted to MOODLE of the link here: [HOW TO GENERATE REVIEW MATERIALS](#)

CH 22

- 1. Historically, who and when did individuals contribute to ideas about what we know as “evolution” today.
 - a. Lamarck vs. Darwin? How did their views of “evolution” differ?
 - b. Who contributed ideas about how living organisms came to exist on Earth before Darwin?
- 2. What exactly specifically were Darwin’s contributions to the area of “evolution”
 - a. What were his research areas?
 - b. How did his ideas differ from others?
 - c. What were his 3 broad observations and how do they tie to evolution?
 - d. How do Artificial Selection, Natural Selection, and Adaptation differ?
- 3. What are the key features of Natural Selection?
- 4. What are the differences between Darwin’s OBSERVATIONS and INFERENCES and what were they?
- 5. What is the world population now? Has it grown or shrunk in recent decades?
- 6. What are the four types of data document the pattern of evolution and examples of each one?
- 7. What is the difference between Anatomical and Molecular Homologies and how can they be used in studying evolution?
 - a. What are vestigial structures?
 - b. Examples of vestigial structures in organisms?
- 8. Understand the difference between Analogous & Homologous Structures
 - a. Be able to compare and contrast organisms for these types of structures.
 - b. Explain why convergent evolution does not provide information about ancestry.
- 9. How can Biogeography be used to help understand evolutionary events.
- 10. What does “endemic” mean?
- 11. You should be able to use or create a DICHOTOMOUS KEY. This is what we did using the different coins and types of balls during the in-class activity.

CH 23

1. How does the process of Natural Selection work?
 - a. Populations?
 - b. Individuals?
2. What are the 3 main mechanisms that can cause allele frequency to change? How does each one work?
3. What is genetic variation and how does it relate to the process of evolution?
 - a. Introns vs. Exons?
 - b. Sex determination mechanisms in various organisms
 - c. What are “mutations” and examples of different types?
 - d. What does the “degeneracy of the genetic code” refer to?
4. Sexual reproduction as it relates to evolution
 - a. three areas of sexual reproduction that contribute to genetic variation
 - b. which area contributes the MOST variation and why/how?
5. Application of Hardy Weinberg (HW)
 - a. You will have an example on the exam where you will need to apply the HW formula and interpret your results.
 - b. Conditions required for HW
 - c. What is PKU and how can we relate it to HW?
 - d. You should be able to apply and interpret the HW formula if given a data set (very similar to the MOODLE quiz). This problem on the exam will be worth a significant number of points.
6. What are the three major factors alter allele frequencies and bring about most evolutionary change?
 - a. Natural selection
 - b. Genetic drift
 - c. Gene flow
 - d. Difference between founder effect and bottleneck effect?
7. What are the are three modes of selection and how do they differ from each other?
8. What is Sexual Selection and how does it relate to evolutionary events?
 - a. sexual dimorphism
 - b. intra-sexual selection
 - c. inter-sexual selection
9. What is Balancing selection?
 - a. Frequency-dependent selection
 - b. Heterozygote advantage as it relates to malaria
10. What does FITNESS measure?
 - a. ABSOLUTE fitness
 - b. RELATIVE fitness
 - c. You should be able to calculate absolute and relative fitness values if given a small data set (see the cat eyes and colored roses example from the notes).

CH 24

1. What are the different ways in which we can define a “species”?
 - a. How do they differ from each other?
 - b. Which one is best/worst?
 - c. Which one is most generally accepted as the most accurate?
2. What is Reproductive isolation and how does it relate to mating success?
 - a. prezygotic barriers and examples

- b. postzygotic barriers and examples
- 3. What is a mule?
 - a. How do you make more mules?
- 4. What are the ways/mechanisms in which speciation can occur?
 - a. Allopatric speciation
 - b. Sympatric speciation
 - c. be able to recognize the 4 different types of speciation
- 5. What is polyploidy and how does it happen?
 - a. autopolyploid?
 - b. allopolyploid?
 - c. what is the difference between TRIPLOID and TRISOMIC?
- 6. What is a hybrid zone and how can these zones occur?
 - a. What is the result of a hybrid zone over time?
 - i. Reinforcement
 - ii. Fusion
 - iii. Stability
- 7. What is punctuated equilibrium and how does it relate to speciation events?
- 8. Difference between Microevolution and Macroevolution?

CH 25

- 1. Characteristics of early Earth?
- 2. Important people/experiments as they relate to examining the physical/chemical states of early Earth
- 3. Why are meteorites important to study as they relate to “life”?
- 4. What is a “protocell” and why are they important to study/understand as they relate to “life”?
- 5. What is a “fossil” and why are they important to study/understand as they relate to “life”?
 - a. Carbon dating?
 - b. Half-life?
 - c. Be able to calculate a “half-life” if given data
- 6. The geologic record is divided into four major eons
 - a. Hadean
 - b. Archaean
 - c. Proterozoic
 - d. Phanerozoic eons
 - e. What are some major events that take place in each eon? (*Be sure to look at SLIDE 20 which lists the 8 majors events as they occurred using the analogy of a clock)
- 7. What is a stromatolite and how old are they?
- 8. When did atmospheric oxygen appear in the fossil record?
- 9. When did the first eukaryotic cells appear in the fossil record?
 - a. serial endosymbiosis?
 - b. how does this process work?
- 10. What is the key evidence supporting an endosymbiotic origin of mitochondria and plastids?
- 11. Importance of the Cambrian “explosion”?
- 12. How do mass extinctions play a role in evolution?
 - a. How many occurred?
 - b. Factors that might have contributed to mass extinction events?
 - c. Consequences of mass extinctions?

13. What is “Adaptive radiation”?
 - a. examples?
 - b. what could cause this process?
14. What do the terms “heterochrony” and “paedomorphosis” refer to?